# DUNE: A Reproducible, Low-Cost Indoor UWB Positioning Stack for Small Rovers

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***Abstract*—This paper presents DUNE, an open-source indoor ultra-wideband (UWB) positioning system for small mobile robots built from commodity DecaWave DW1000 transceivers. The system integrates a two-stage range calibration that removes antenna-delay and received-power biases, a continuous non-line-of-sight (NLOS) weighting derived from the first-path-to-total received-power ratio, and a moving-horizon estimator that exploits a two-tag rigid-body constraint to observe heading. Anchor coordinates are determined automatically from inter-anchor ranging, eliminating any external survey of the infrastructure. As the estimator depends only on standard range and first-path-power reports, it applies without modification to successor transceivers such as the DW3000. The system is evaluated on a five-anchor sand-arena testbed against an overhead-camera reference: calibration reduces the single-reading horizontal error from $199$ to $94\,\mathrm{mm}$, temporal filtering attains $60$–$65\,\mathrm{mm}$ at rest, the self-survey recovers the anchor geometry to $51\,\mathrm{mm}$, and a driven trajectory under continuous ground truth is tracked at $124\,\mathrm{mm}$ RMSE. The complete firmware, host software, calibration data, and logs are released.**

***Index Terms*—Ultra-wideband, indoor localization, range sensing, NLOS mitigation, sensor fusion, calibration, reproducible robotics.**

## I. Introduction

Accurate indoor localization is a prerequisite for autonomous mobile robots operating where satellite navigation is unavailable. Among the candidate technologies, ultra-wideband (UWB) ranging offers a favourable combination of centimetre-level time-of-flight resolution, robustness to multipath relative to narrowband radio, and low unit cost, and it has become a common choice for real-time localization systems [1]. A single DW1000 transceiver can time a radio round trip to a few centimetres, and the underlying two-way ranging and multilateration are well established [2].

In practice, however, a substantial gap separates this nominal ranging resolution from the pose accuracy a robot experiences during operation. Raw transceiver ranges carry hardware- and signal-dependent biases of several decimetres; obstructed, non-line-of-sight (NLOS) paths inflate the measured distance without any external indication; and the anchor coordinates that anchor the whole solution must themselves be obtained, conventionally by manual survey or an external camera on which the deployment then depends. Closing this gap is largely a matter of engineering that is seldom reported in reproducible detail, so each new deployment tends to repeat it.

This paper documents a complete, low-cost UWB positioning stack, DUNE, that addresses these issues end to end and is released in full. The system localizes a small Ackermann rover from five fixed DW1000 anchors and one or two ESP32-hosted tags, with all estimation performed on a host computer (Fig. 1). The design targets reproducibility on commodity hardware rather than a single algorithmic novelty, and its accuracy derives from the disciplined combination of calibration, NLOS-aware estimation, and automatic infrastructure localization described in the following sections.

The main contributions are:

1) A complete and openly released pipeline from raw double-sided two-way ranging to a filtered vehicle pose, including a two-step radio calibration (antenna delay and received-power range bias). Unlike common open DW1000 libraries, which target a fixed small anchor set, the pipeline supports an arbitrary number of anchors ($N \geq 4$) and degrades gracefully when only a subset responds.
2) A continuous first-path-power NLOS weighting integrated with a robust weighted least-squares solver and a moving-horizon estimator that enforces a two-tag rigid-body constraint to recover heading.
3) An automatic anchor self-localization procedure that recovers the anchor geometry from inter-anchor ranging alone, eliminating the need for an external survey of the infrastructure.

## II. Related Work

### A. UWB positioning and NLOS mitigation

UWB indoor positioning has been studied extensively; a comprehensive survey is given in [1]. Fully open systems remain comparatively rare: ATLAS provides an open time-difference-of-arrival localizer [3], and the UTIL dataset [4] releases labelled ranging logs for benchmarking. A dominant error source in cluttered environments is non-line-of-sight propagation, in which an obstructed direct path forces the signal along a longer route and biases the range positively. Kolakowski and Modelski [5] mitigate this by classifying each link as line-of-sight or NLOS from the DW1000 first-path power and feeding the decision to an extended Kalman filter. Rather than a hard classification, we derive a continuous weight from the first-path-to-total power ratio measured within the same range report, which both attenuates suspect ranges in the position solve and inflates their variance in the temporal filter.

Manuscript received [TODO: date]; revised [TODO: date].

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Code, calibration data, and logs: https://github.com/AdershM-2/UWB_3.

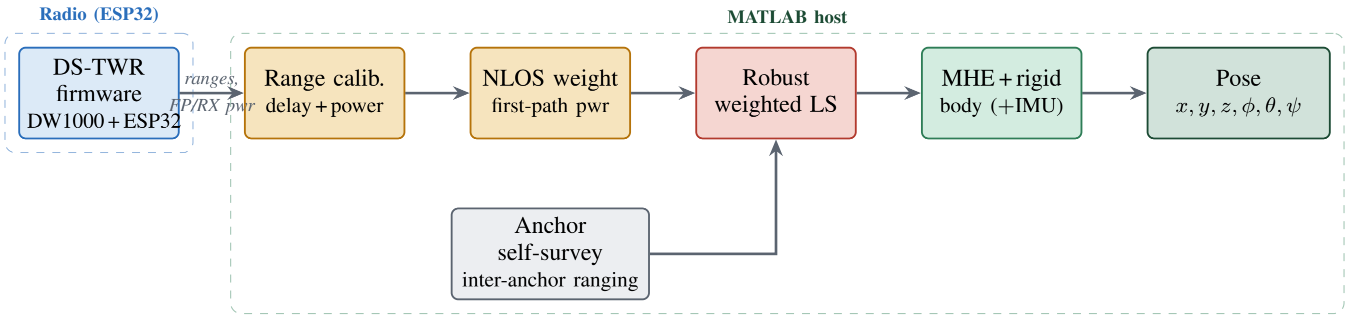


Fig. 1. The DUNE pipeline. Five DW1000 anchors and one or two ESP32 tags perform double-sided two-way ranging; the host calibrates each range for antenna delay and received-power bias, weights it by a first-path NLOS score, solves a robust least-squares fix, and fuses the stream in a moving-horizon estimator with a two-tag rigid-body constraint. Anchor coordinates are supplied by an inter-anchor self-survey, and roll, pitch, and height are read from the on-board IMU to complete the six-degree-of-freedom pose.

### B. Anchor self-calibration

The anchor coordinates required by any multilateration system are conventionally measured by hand or with external instrumentation. Self-calibration methods that instead infer the anchor layout from radio measurements are surveyed in [6]. Recent work scales anchor calibration to large environments using Gaussian processes, but relies on a LiDAR-inertial odometry trajectory as input [7]. In contrast, the self-survey used here requires only the pairwise ranges the anchors can already measure among themselves, and recovers the constellation by classical multidimensional scaling [8].

## III. Problem Definition

Consider a set of $N \geq 4$ fixed anchors with positions $\{\mathbf{a}_k\}_{k=1}^{N} \subset \mathbb{R}^3$ expressed in a common world frame. A rover carries $M \in \{1, 2\}$ tags on a rigid plate; for $M{=}2$ the two tag antennas are separated by a known baseline $L$. The rover motion is planar, so its configuration is the body pose $\mathbf{x} = (c_x, c_y, \psi) \in SE(2)$—planar position $\mathbf{c}$ and heading $\psi$—which is extended to the full six-degree-of-freedom pose using the height $z$ and the roll and pitch $(\phi, \theta)$ read from the on-board IMU. The two tag positions follow from the body pose as

$$\mathbf{p}_{1,2} = \mathbf{c} \pm \tfrac{L}{2}\,[\cos\psi,\ \sin\psi]^{\top}, \tag{1}$$

for the front and rear tag respectively.

Each double-sided two-way ranging exchange between tag $i$ and anchor $k$ produces a range estimate that departs from the true geometric distance by several additive terms,

$$\rho_{ik} = \|\mathbf{p}_i - \mathbf{a}_k\| + \delta_k + \gamma_k(P_{ik}) + \Delta_{ik} + \varepsilon_{ik}, \tag{2}$$

where $\delta_k$ is the per-anchor antenna-delay bias, $\gamma_k(P_{ik})$ a bias that depends on the received power $P_{ik}$, $\Delta_{ik} \geq 0$ the excess path length introduced under NLOS conditions, and $\varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2)$ zero-mean measurement noise. The tags report at different instants, so the ranges arrive asynchronously rather than as synchronized snapshots.

The problem is twofold. First, given the anchor positions and a stream of asynchronous ranges (2), estimate the body trajectory $\{\mathbf{x}_t\}$ online at the sensor rate ($\geq 5\,\mathrm{Hz}$) while suppressing the systematic biases and NLOS excess. Second, when the anchor positions are not externally available, estimate $\{\mathbf{a}_k\}$ from inter-anchor ranging alone. The remainder of the paper addresses these two problems and evaluates the resulting accuracy against an independent optical ground truth.

## IV. Methodology

### A. Ranging and range calibration

Each range is obtained by a double-sided two-way exchange (DS-TWR), whose symmetric four-timestamp formulation cancels the clock-frequency offset between the two uncoordinated crystals to first order [2] (Fig. 2). One DW1000 timer tick corresponds to $4.69\,\mathrm{mm}$ of range and sets the resolution of every calibration constant below. The two systematic biases in (2) are removed in sequence. The antenna-delay term $\delta_k$, which reflects the propagation time internal to each board, is calibrated per anchor against a reference position and stored on the device, after which all anchors agree with the reference distance to within $\pm 16\,\mathrm{mm}$. The residual power-dependent bias $\gamma_k$ is modelled per anchor as an affine function of the first-path power $P_{\mathrm{fp}}$, which is more stable than total power,

$$\hat{d}_k = \rho_k - \big(a_k\, P_{\mathrm{fp},k} + b_k\big), \qquad a_k \in [7.7,\ 9.5]\ \mathrm{mm\,dB^{-1}}, \tag{3}$$

with $P_{\mathrm{fp}}$ clamped to the fitted interval so that the model is never extrapolated. The coefficients are estimated once by leave-one-position-out regression over a static campaign.

### B. Robust NLOS-aware estimation

Under obstruction the first signal component is attenuated relative to the total received energy, so the gap between total and first-path power is a direct indicator of the NLOS excess $\Delta_{ik}$ (Fig. 3). Denoting this gap $g_k = P_{\mathrm{rx},k} - P_{\mathrm{fp},k}$ in decibels, each range is assigned a continuous weight

$$w_k = \max\Big(w_{\mathrm{f}},\ 10^{-\max(0,\, g_k - \tau)/10}\Big), \tag{4}$$

with LOS threshold $\tau = 3\,\mathrm{dB}$ and floor $w_{\mathrm{f}} = 0.05$: below the threshold a range keeps unit weight, above it the weight decays with the excess gap, and the floor prevents any single anchor from being discarded outright. The instantaneous fix follows from robust weighted multilateration,

$$\mathbf{p}^{\star} = \arg\min_{\mathbf{p}} \sum_{k=1}^{N} w_k\, \rho_\delta\big(\|\mathbf{p} - \mathbf{a}_k\| - \hat{d}_k\big), \tag{5}$$

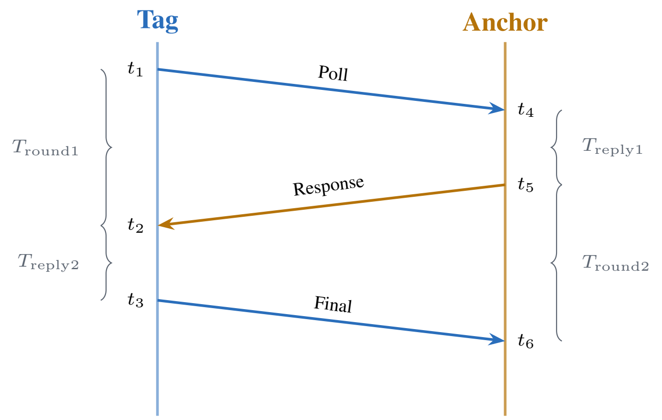


Fig. 2. Double-sided two-way ranging. Four host-side timestamps over two round trips recover the time-of-flight without a shared clock; the symmetric form is first-order insensitive to crystal frequency offset.

where $\rho_\delta$ is the Huber loss and a $\chi^2$ consistency gate rejects gross outliers, so that an inconsistent range bends rather than dominates the solution.

The trajectory itself is estimated by a moving-horizon estimator (MHE) that, at each step, jointly optimizes the last $N_\text{h}$ states ($N_\text{h}$=10, $\approx$2 s) so that new evidence can revise the recent estimate:

$$\begin{aligned}\min_{\{\mathbf{x}_k\}} \sum_k \sum_j w_{kj}\, \rho_\delta\big(\|\mathbf{p}_k - \mathbf{a}_j\| - \hat{d}_{kj}\big) \\ + \sum_k \big\|\mathbf{x}_{k+1} - f(\mathbf{x}_k)\big\|^2_{Q^{-1}} + \big\|\mathbf{x}_1 - \bar{\mathbf{x}}\big\|^2_{P^{-1}},\end{aligned} \tag{6}$$

where the first term fits every windowed range under the same weighted robust loss, the second penalizes departures from the kinematic motion model $f(\cdot)$, and the third is an arrival cost coupling the window to the previous estimate. The same weights $w_{kj}$ scale the measurement variances, so the NLOS score governs both the fix and the filter. The IMU contributes only differential quantities—yaw rate and gravity-referenced roll and pitch—never an absolute heading; because the Ackermann platform cannot slip laterally, $f(\cdot)$ constrains the body velocity to the heading direction. For two tags, the rigid constraint (1) is imposed by construction, so a single body state generates both tag predictions, the baseline is exact in every solution, and the heading is observed directly by the radios with a per-reading resolution $\sigma_\psi \approx \sqrt{2}\,\sigma_d/L$ ($\approx 4.6°$ for $\sigma_d = 30$ mm, $L = 0.527$ m), further reduced by filtering.

### C. Automatic anchor localization

When the anchor coordinates are not measured externally, they are recovered from the ten pairwise inter-anchor ranges of the five-anchor set. A set of mutual distances defines a rigid configuration up to an isometry, and classical multidimensional scaling [8] returns the point set whose pairwise distances best match the measured ranges. The triggering tag only relays messages, so its own antenna delay does not enter the survey. The recovered constellation is determined up to a rotation and reflection, resolved against a single known landmark or, for evaluation, aligned to a reference layout without scaling.

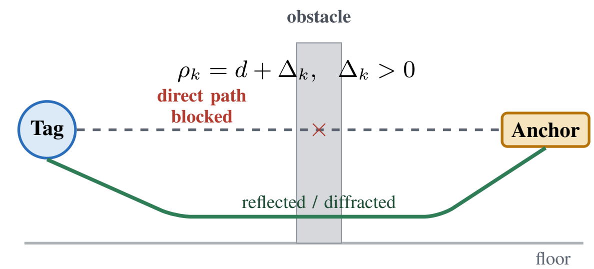


Fig. 3. NLOS mechanism. When the direct path is obstructed the signal reaches the anchor along a longer reflected or diffracted route, so the measured range is biased positively by $\Delta_k$. The bias is estimated from the total-to-first-path power gap and used to weight the range through (4).

TABLE I
STATIC SINGLE-READING HORIZONTAL ERROR OVER 19 POSITIONS.

| | **RMSE** | **Median** | **95th pct.** |
|---|---|---|---|
| Antenna-delay calibration only | 199 | 132 | 393 |
| + received-power correction | **94** | **69** | 160 |
| Oracle (per-position constant) | 72 | 31 | — |

All values in mm.

## V. Experimental Results

The system was evaluated in a sand-filled arena with five DW1000 anchors—four at ground level near the corners and one elevated—whose coordinates were fixed either by the self-survey of Section IV or by an overhead camera. The rover carried a front and a rear tag separated by $L = 0.527$ m, the rear tag additionally equipped with a BNO085 IMU, together with an AprilTag fiducial facing the ceiling. A ceiling-mounted Kinect v2 tracked the fiducial for ground truth; after intrinsic calibration and clicked-point registration to the anchor frame the registration residual was 13.5 mm, and the marker lever arm was compensated per sample. All estimation was performed off-board.

### A. Static accuracy and calibration

Localization accuracy was first characterized with the rover stationary at 19 surveyed positions across the arena ($\sim$20 s per position, $\sim$ 7 600 ranges). Table I reports the single-reading horizontal error under the two calibration stages. Antenna-delay calibration alone leaves a floor-wide RMSE of 199 mm; adding the received-power correction of (3) reduces it to 94 mm (Fig. 4), the single largest improvement in the pipeline.

To quantify how much of the remaining error is irreducible by any static correction, we evaluated an oracle that subtracts, from each position, the mean residual measured there—the best achievable by any time-invariant, position-dependent correction. Its residual RMSE is 72 mm, which therefore lower-bounds any fixed calibration and shows that the leftover error is not a static spatial map that a denser survey could remove. Temporal filtering nonetheless attains 60–65 mm at rest, below this bound, because it averages the zero-mean high-frequency component of the residual over successive readings, leaving only a slowly varying multipath term that the static oracle cannot capture.

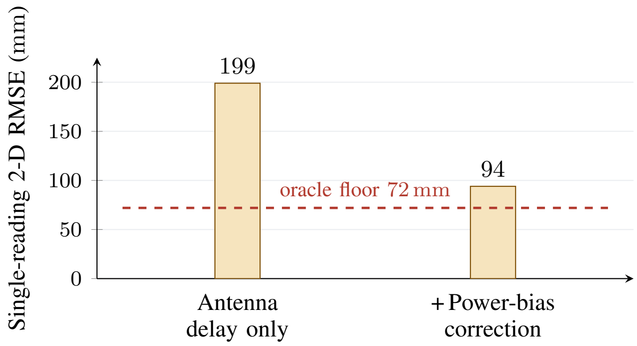


Fig. 4. Effect of the two-step calibration on floor-wide single-reading error. The received-power correction removes the strength-dependent residual left by antenna-delay calibration; the dashed line marks the 72 mm oracle floor of any fixed correction.

TABLE II
MOVING-HORIZON ESTIMATOR VERSUS KALMAN FILTER (SAME LOGS).

| Metric | Kalman (EKF) | MHE |
|---|---|---|
| Trajectory jerk, moving ($\mathrm{m\,s^{-3}}$) | 0.61 | **0.36** |
| Additional lag (mm) | — | 6 |
| RMSE at rest (mm) | 64 | **61** |
| Solve time (ms) | $< 1$ | 3 |

### B. Estimator comparison and anchor self-survey

Table II compares the moving-horizon estimator with an extended Kalman filter on identical recorded ranges. While the vehicle is moving, the MHE reduces trajectory jerk by 40–44 % (from 0.61 to $0.36\,\mathrm{m\,s^{-3}}$) for only 6 mm of additional lag, and the two estimators are equivalent at rest; each MHE solve completes in 3 ms, within the 6 Hz update budget. The reduced jerk reflects the window optimization rejecting isolated range excursions that a single-step filter propagates.

The anchor self-survey was evaluated against the independently registered overhead-camera layout (Table III). The raw inter-anchor ranges are biased long by up to 202 mm (RMS 118 mm) because each measurement carries the antenna delays of both participating anchors. After multidimensional scaling and alignment without scaling, the reconstructed constellation agrees with the camera layout to 51 mm RMS once the per-anchor delay bias is removed—an independent confirmation of the geometry to within centimetres. The deployed configuration is built from this self-survey.

### C. Trajectory tracking

The full system was finally evaluated on a driven run under continuous optical ground truth. Over the trajectory the Kinect provided 133 valid pose samples at near-continuous coverage, and both tags were serviced evenly (355 and 362 sweeps) with an average of five anchors per sweep, so the ground truth constrains the entire path rather than isolated segments. The rigid-body MHE track follows the reference with a horizontal RMSE of 124 mm; the estimated path length is $1.4\times$ that of the truth, indicating that the residual is dominated by high-frequency wander rather than a systematic offset, as the two trajectories share the same overall shape. This moving accuracy is consistent with the single-reading static scatter accumulating under motion, and is limited by per-sweep range noise, which longer averaging windows trade against lag; reducing it further through tighter range gating and higher sweep rates is the subject of ongoing work.

TABLE III
ANCHOR SELF-SURVEY VERSUS OVERHEAD-CAMERA LAYOUT.

| Quantity | Value (mm) |
|---|---|
| Raw inter-anchor range bias (max) | 202 |
| Raw inter-anchor range bias (RMS) | 118 |
| Reconstructed geometry vs. camera (RMS) | **51** |

## VI. CONCLUSION

We presented DUNE, a complete and openly released indoor UWB positioning stack for a small mobile robot. Its accuracy follows from a two-step range calibration, a continuous first-path-power NLOS weighting, and a moving-horizon estimator with a two-tag rigid-body constraint, while the anchor geometry is obtained automatically from inter-anchor ranging rather than external survey. The system achieves a filtered static accuracy of 60–65 mm and reconstructs the anchor layout to 51 mm of an optical reference. Because it operates only on standard range and first-path-power reports, the pipeline transfers without modification to newer transceivers such as the DW3000. Ongoing work addresses the moving-accuracy gap through tighter range gating and higher sweep rates. All firmware, host software, calibration data, and logs are released to support reproduction and extension.

## VII. IMPLEMENTATION AND REPRODUCIBILITY

The complete system is released at https://github.com/AdershM-2/UWB_3. The host stack is a MATLAB package (`+dune`) with supporting scripts (Table IV); the firmware comprises three Arduino sketches for the anchor and the two tag variants. The ingest layer is protocol-agnostic—a sweep is a set of (anchor, range, quality) samples—so the estimators consume live serial data and replayed logs identically, and estimation is kept out of the real-time driving loop, which only records. The two calibration files that reproduce the accuracy are `delay_calibration.json` (per-anchor antenna delays, in 4.69 mm ticks) and `range_correction.json` (the per-anchor first-path-power coefficients of (3)); both load automatically in `solveSweep`. The static-campaign readings, driven-run recordings, and an extended engineering report are released alongside the code.

## REFERENCES

[1] A. Alarifi, A. Al-Salman, M. Alsaleh, A. Alnafessah, S. Al-Hadhrami, M. A. Al-Ammar, and H. S. Al-Khalifa, "Ultra wideband indoor positioning technologies: Analysis and recent advances," *Sensors*, vol. 16, no. 5, p. 707, 2016.

[2] D. Neirynck, E. Luk, and M. McLaughlin, "An alternative double-sided two-way ranging method," in *Proceedings of the 2016 13th Workshop on Positioning, Navigation and Communications (WPNC)*, 2016, pp. 1–4, tODO: verify page numbers.

TABLE IV
KEY RELEASED SOFTWARE MODULES.

| **Module** | **Function** |
|---|---|
| `Anchor.ino`, `Tag*.ino` | DW1000 DS-TWR firmware (anchor and two tag variants). |
| `+dune/solveSweep` | Calibrate, weight, and solve one sweep. |
| `+dune/multilaterate` | Robust weighted least-squares solver, (5). |
| `+dune/gapWeights` | First-path-power NLOS weights, (4). |
| `+dune/MheEstimator` | Moving-horizon estimator (CasADi/IPOPT), (6). |
| `+dune/MheRigid` | Two-tag rigid-body estimator. |
| `tune_delays_sweep.m` | Over-the-air antenna-delay calibration. |
| `survey_reconstruct.m` | Multidimensional-scaling anchor self-survey. |
| `rover_run_report.m` | Driven-run scoring against Kinect truth. |

[3] J. Tiemann, F. Schweikowski, and C. Wietfeld, “ATLAS: An open-source TDOA-based ultra-wideband localization system,” in *2016 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2016, tODO: verify page numbers.

[4] W. Zhao, A. Goudar, X. Qiao, and A. P. Schoellig, “UTIL: An ultra-wideband time-difference-of-arrival indoor localization dataset,” *The International Journal of Robotics Research*, vol. 43, no. 10, pp. 1443–1456, 2024.

[5] M. Kolakowski and J. Modelski, “First path component power based NLOS mitigation in UWB positioning system,” in *Proceedings of the 25th Telecommunications Forum (TELFOR)*, Belgrade, Serbia, 2017, also available as arXiv:2403.19706. TODO: verify page numbers and DOI.

[6] M. Ridolfi, A. Kaya, R. Berkvens, M. Weyn, W. Joseph, and E. De Poorter, “Self-calibration and collaborative localization for UWB positioning systems: A survey and future research directions,” *ACM Computing Surveys*, vol. 54, no. 4, 2021, article 77, published April 2021.

[7] S. Yuan, B. Lou, T.-M. Nguyen, P. Yin, M. Cao, X. Xu, J. Li, J. Xu, S. Chen, and L. Xie, “Large-scale UWB anchor calibration and one-shot localization using Gaussian Process,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025, also available as arXiv:2412.16880.

[8] W. S. Torgerson, “Multidimensional scaling: I. theory and method,” *Psychometrika*, vol. 17, pp. 401–419, 1952.